CP373 Ethics Reflection Journal

Introduction

I will discuss the ethical consequences of algorithmic transparency and bias in the atmosphere of computing, in particular, how automated systems may contribute to the strengthening of socioeconomic inequalities without any intent. Over the course, I have learned that ethics is not a peripheral obstacle that needs to be overcome once the development has been completed, but a fundamental aspect of responsible engineering. One of the most influential questions that has piqued my thinking is the question of whether a black box algorithm can ever be neutral, or whether the objective to achieve raw efficiency will always cover up systemic prejudice. Based on the philosophical perspectives of Aristotle and Thomas Hobbes, I will state that the technology professional's responsibility needs to be changed to the virtue ethical approach. The personality of the developer and the level of his adherence to transparency, in this perspective, is as important to the good of society as the technical performance of the code itself. The reflection is used to provide a gap between the abstract reasoning of moral theory and the technical issues of modern software practitioners.

Reflection 1: Critical Engagement with a Course Theme

I was particularly interested in the theme of algorithmic bias, as it points out the logical fallacy that data driven systems are objective in nature. When it comes to software development, the training sets are often accepted as the new reality, but they often reflect the historical imbalances that, once automated, create self-reinforcing discrimination loops. According to the utilitarian view as described by John Stuart Mill, an action is morally right when it leads to maximum happiness for the majority.

Nevertheless, when a financial algorithm has an overall high accuracy rate, but the minority group is systematically disadvantaged by it, it would not pass the utilitarian test since it generates a focalized and unfair disutility to this group of people. This struggle brings out the point that naked efficiency may at times be an enemy of real ethical utility. An alternative view is presented by Thomas Hobbes, who considers the absence of ethical regulation in tech as a possible state of nature where the unregulated chase after corporate gain results in a war of all against all. In this kind of world, where a social contract or moral governance does not exist, the authority of algorithms can be predatory and life-threatening to the underprivileged. This is supported by Onibokun et al. (2023) analysis, which states that black box models of financial decision making do not provide the explainability necessary to enable the user to interpret the result or dispute it. This is non-transparency, and it contravenes the fundamental principle of fairness in a society. Combining these theories, I have come to understand that I am not only supposed to optimize, be it speed or accuracy, but also act as a gatekeeper to codified bias. We should stop perceiving ethics as a compliance questionnaire and view it as a tight technical standard. According to a scholarly study by the World Journal of Advanced Research and Reviews (2025), algorithmic hygiene should be a proactive initiative where the developers of algorithms should audit their model to identify disparate impact before implementing the system in a setting where it can do widespread harm. It is only through recognition of these power relationships that we will be able to create systems which are genuinely of the common good.

Reflection 2: Real-World Application

A striking real world application of these ethical challenges is the rise of AI powered business analytics and credit scoring. Financial institutions increasingly rely on non-traditional data, such as digital footprints, social media behaviour, or even geographic location to determine an individual's

creditworthiness. While technically sophisticated and highly efficient, this practice often results in a modern form of "digital redlining," where proxy variables for race or socio economic status lead to the systemic exclusion of qualified individuals from the global financial system. This scenario perfectly illustrates the profound difficulty of professional responsibility: a developer may write "clean" bug free code that is logically sound yet socially destructive. Applying Aristotle's concept of *phronesis*, or practical wisdom, we see that a virtuous professional must look beyond the immediate technical task to understand the broader human impact of their creations. Aristotle argues that "good" is achieved when a thing performs its function excellently in accordance with virtue, suggesting that technical skill alone is insufficient for true excellence. In the context of computing, this means that a "good" algorithm is one that is not only mathematically accurate but also fundamentally just and fair. Recent findings from the World Journal of Advanced Research and Reviews (2025) suggest that without robust ethical governance and human-centric design, these automated tools can exacerbate existing financial inequalities. This controversy highlights that technical "optimizations" are never truly value-neutral; they carry the biases of their environment. As a Computer Science student at Laurier, this case study serves as a vital warning that my future work in financial modelling could have life-altering consequences for others. Professional responsibility, therefore, demands an unwavering commitment to transparency and explainability, ensuring that the logic behind a financial decision is clear to those it affects. By adopting this mindset, I can move from being a mere coder to an ethical practitioner who recognizes that the integrity of our digital infrastructure depends entirely on the fairness of its underlying algorithms.

Reflection 3: Personal Position or Ethical Dilemma

Consider a hypothetical dilemma where I am tasked with developing a predictive model for a fintech firm that uses "alternative data", such as social media engagement or shopping habits to predict loan defaults, knowing the data likely contains deep rooted socioeconomic biases. A manager might argue from a narrow utilitarian view that the model is "good" because it increases company profits and correctly identifies 90% of high-risk borrowers, thus maximizing the utility for the majority of the firm's stakeholders. However, my judgment has shifted toward a Virtue Ethics framework, which asks: what kind of person am I becoming by building a system that I know is inherently unfair?. I believe a virtuous developer must prioritize integrity and justice over raw model performance or corporate gain. If a model achieves accuracy by exploiting systemic bias, it is a "vicious" tool rather than a "virtuous" one, as it fails to aim at the "good" of the human community. A possible objection is that businesses must remain competitive and that "perfect," unbiased data simply does not exist in the real world. While this is technically true, it does not absolve the developer of the moral responsibility to mitigate harm. The complexity lies in the fact that once bias is "baked into" a model, it becomes a self-fulfilling prophecy, denying opportunities to those the model has already unfairly labelled as "high risk". Re-evaluating this scenario through the lens of Thomas Hobbes, I recognize that without professional standards that mandate transparency, the technology industry risks losing public trust and collapsing into a chaotic "state of nature". My position is that I would refuse to deploy such a model without first implementing significant debiasing measures and providing a clear, explainable logic of the decision making process to the end user. This course has taught me that my technical skills are a form of power, and using that power responsibly means having the courage to challenge unethical requirements, even when they are profitable or requested by superiors. By standing firm, I uphold the "social contract" of a professional computer scientist.

Conclusion

Reflecting on these themes has deepened my understanding that technical proficiency is hollow without a strong, consistent ethical foundation. I have learned that the philosophical frameworks provided by Aristotle, Hobbes, and Mill are not just historical artifacts but essential tools for navigating the modern dilemmas of AI and big data. These reflections have transformed my view of software engineering from a purely mathematical discipline into a social practice with significant moral weight. Moving forward, this knowledge will inform my career by compelling me to advocate for "ethics by design" in every project I undertake, particularly in the realm of financial modelling. I now recognize that my responsibility as a computer scientist is to ensure that the digital world we build does not replicate the prejudices of the past, but instead fosters a more just and transparent future. Ultimately, the true measure of my professional success will not be the complexity of my code, but the fairness of its impact on society. This commitment to ethical excellence is what will define my contribution to the tech industry.

Works Cited

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